

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Before working on the fuel system, clean the area thoroughly

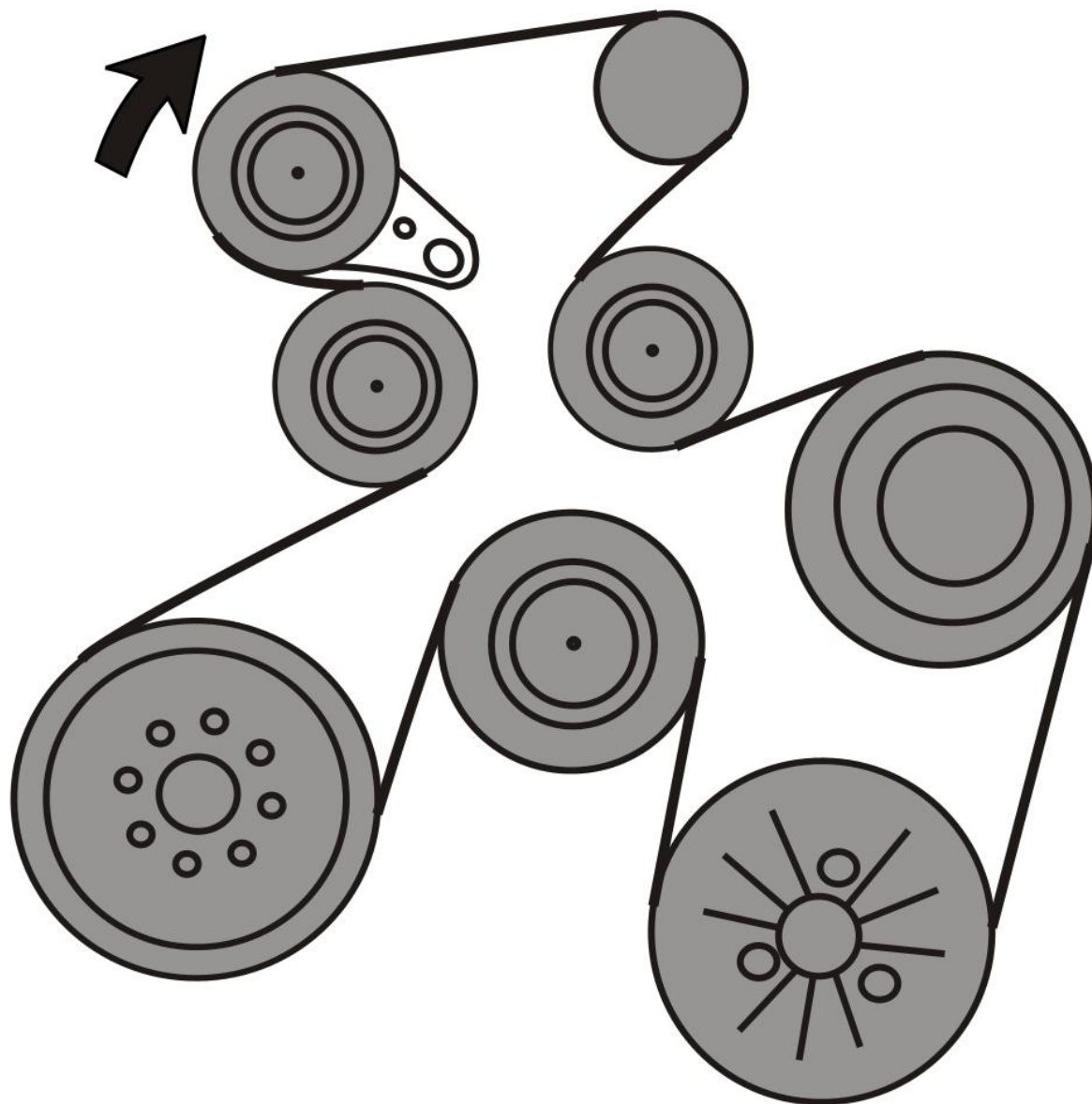
Always use new gaskets

Removal

Disconnect the battery earth cable

Remove the ancillary drive belt

Drive belt layout (With air conditioning)



Remove the dust and pollen filter cover

Remove the engine side cover. On both sides.

Disconnect the hose

Remove the dust and pollen filter housing

Remove the engine upper cover

Remove the ignition coils

Remove the spark plugs

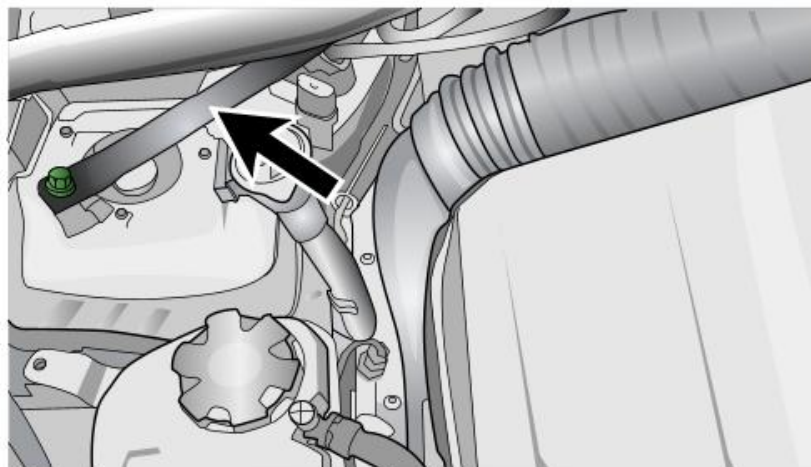
Disconnect the injector wiring

Repair Manuals

Remove the injector lines

Remove the injectors

Remove the strut brace. On both sides.



Disconnect the vacuum hose from the valve cover ((x4))

Remove the valve cover

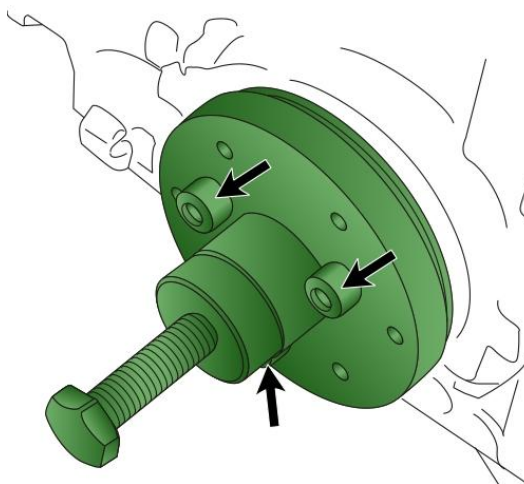
Remove the chain tensioner

Remove the engine lower cover

Remove the crankshaft pulley

Fit the special tools (11 9 221, 11 9 222, 11 9 224)

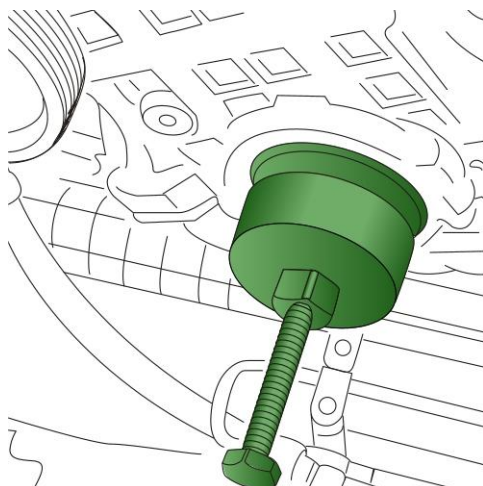
Tighten the special tool ((20 Nm))



Remove the special tools (11 9 221, 11 9 222, 11 9 224)

Fit the special tools ((80 Nm)) (11 0 371, 11 0 372)

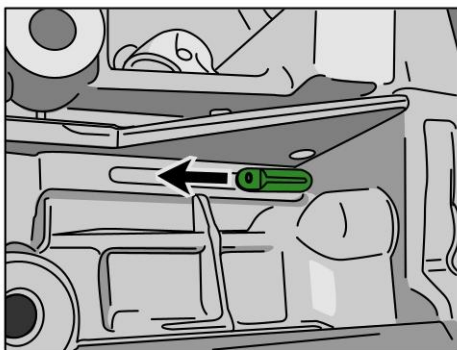
Remove and discard the crankshaft oil seal

Repair Manuals

Remove the tensioner locking pin (11 3 340)

Remove the ancillary drive belt tensioner

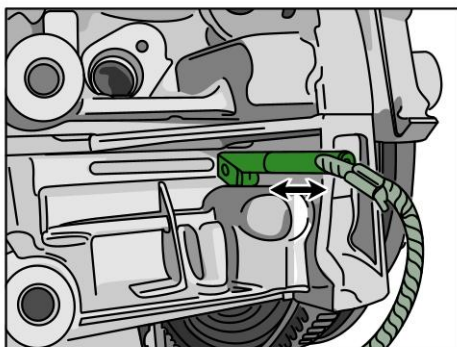
Remove the access plug



Turn the crankshaft to TDC for cylinder 1

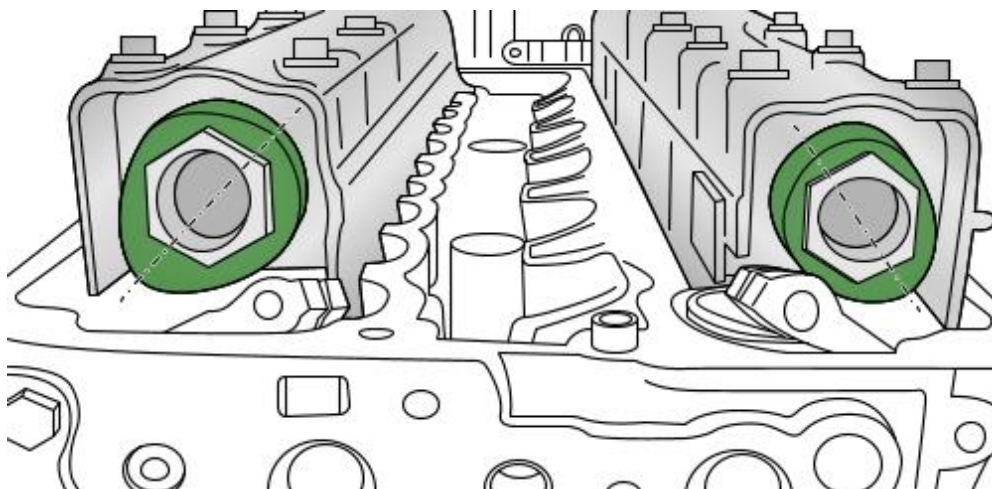
Fit the crankshaft TDC locking pin (11 0 300)

Note: Ensure correct fitting

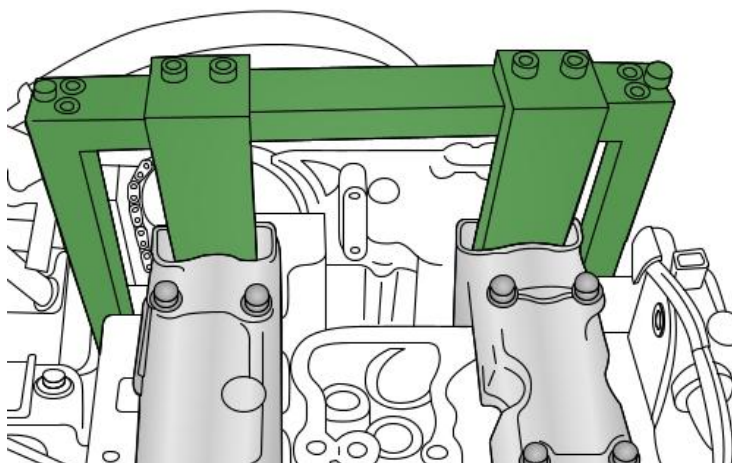


Check the position of the camshafts

Note: Use a mirror

Repair Manuals

Fit the camshaft locking tools (11 4 283, 11 4 281)

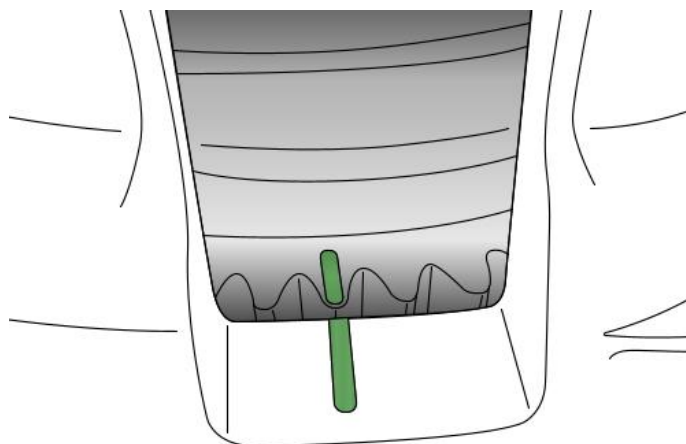


Loosen the camshaft gearwheel bolts

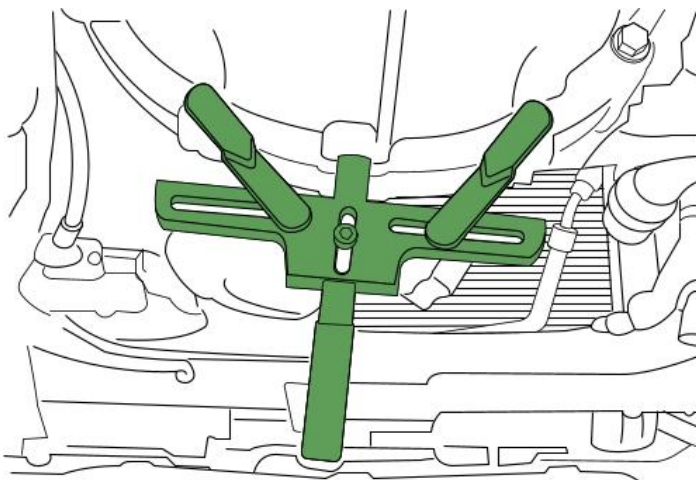
Note: Use a wrench to counterhold the camshafts

Remove the camshaft gearwheels

Note: Mark the position



Fit the crankshaft locking tool (11 8 180)

Repair Manuals

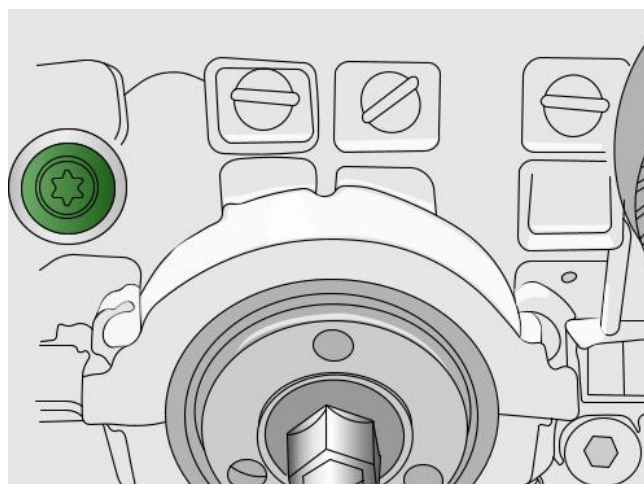
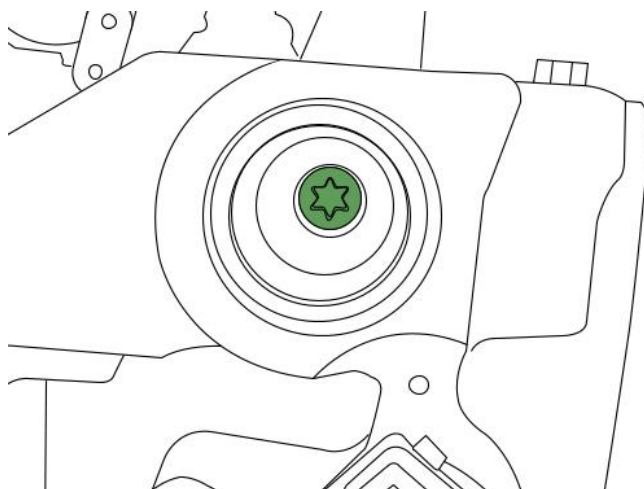
Remove the crankshaft TDC locking pin (11 0 300)

Remove the crankshaft gearwheel centre bolt

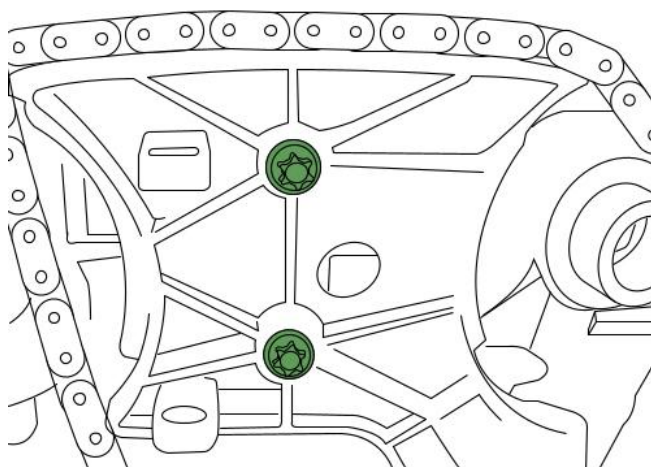
Remove the crankshaft hub

Remove the access plugs

Remove the bearing pins for the chain guide rail and the tensioner rail



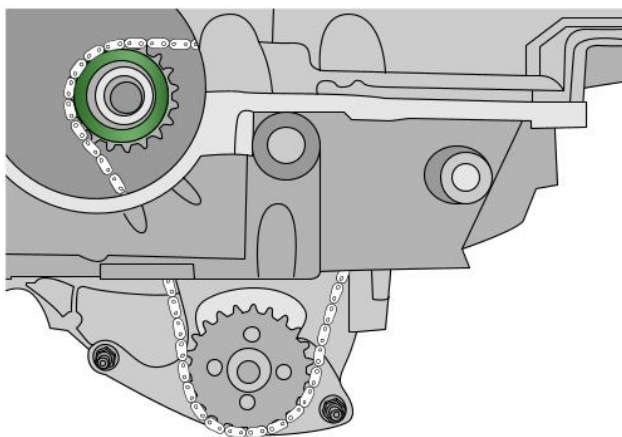
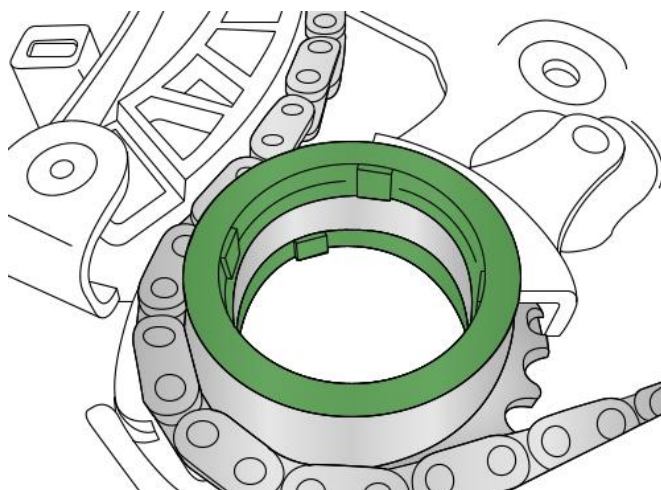
Remove the timing chain guide bolts from the cylinder head

Repair Manuals

Remove the chain guide with the timing chain, guide rail and tensioner rail

Installation

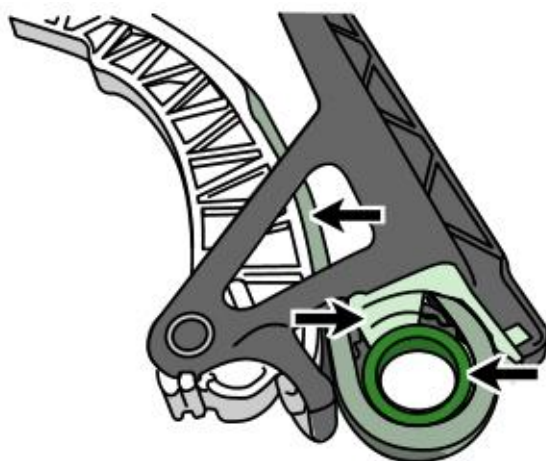
Renew the friction discs



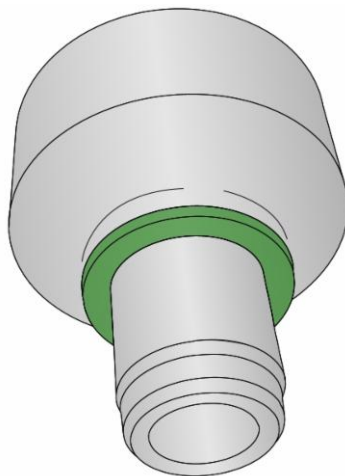
Refit the chain guide with the timing chain, guide rail and tensioner rail

Check for correct fitting

Tension the chain

Repair Manuals

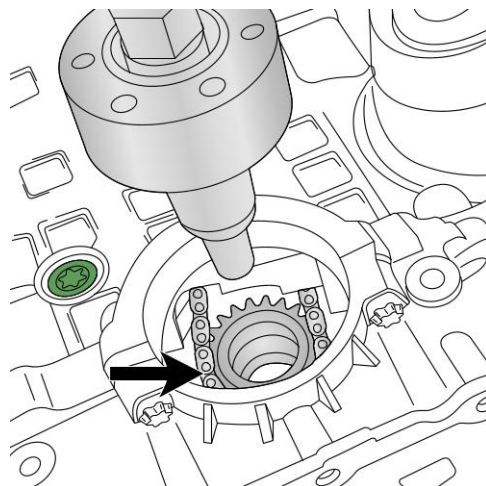
Refit the friction washer



Refit the chain tensioner rail bolt

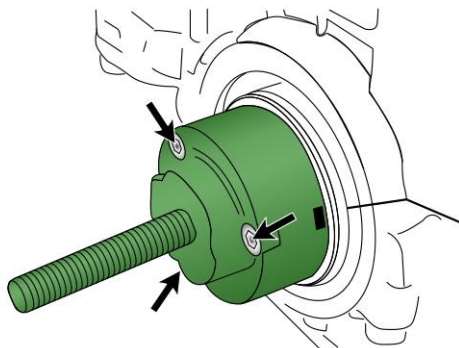
Refit the crankshaft hub

Check for correct fitting



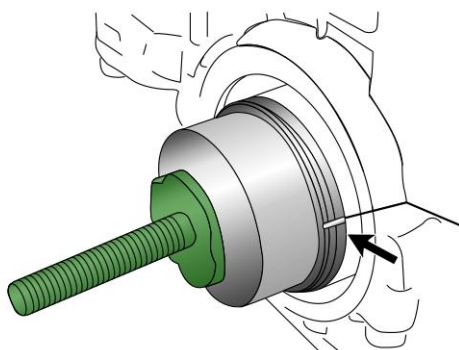
Tighten the crankshaft pulley centre bolt (00 9 140)

Fit the special tools (11 9 231, 11 9 232, 11 9 233, 11 9 234, 11 9 232)

Repair Manuals

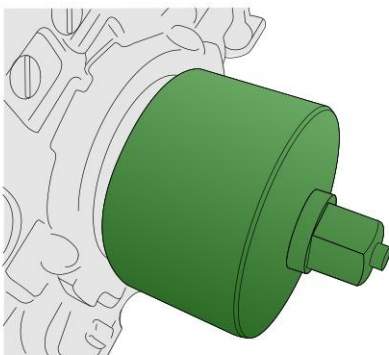
Fit the crankshaft seal over the special tool

Note: The seal should be aligned with the timing cover



Refit the crankshaft front radial seal

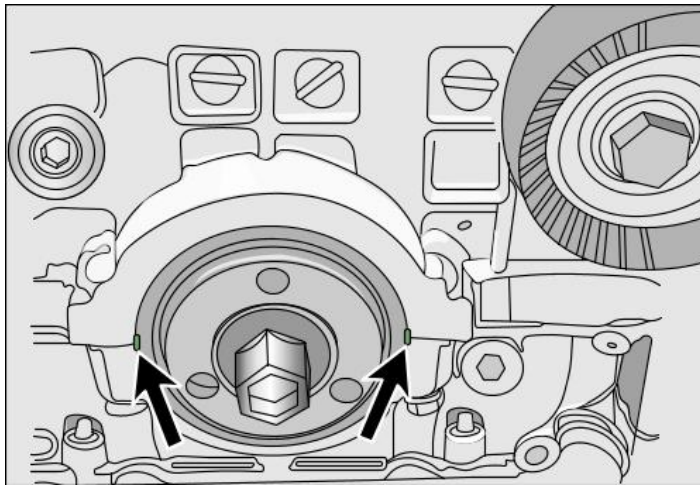
Use the special tool (11 9 231, 11 9 232, 11 9 233, 11 9 234)



Remove the special tools (11 9 231, 11 9 232, 11 9 233, 11 9 234)

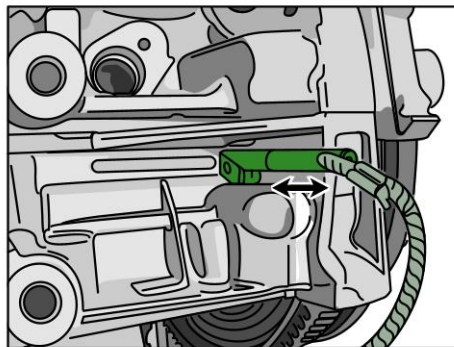
Apply sealant as indicated. Loctite: ((171000)).

Repair Manuals



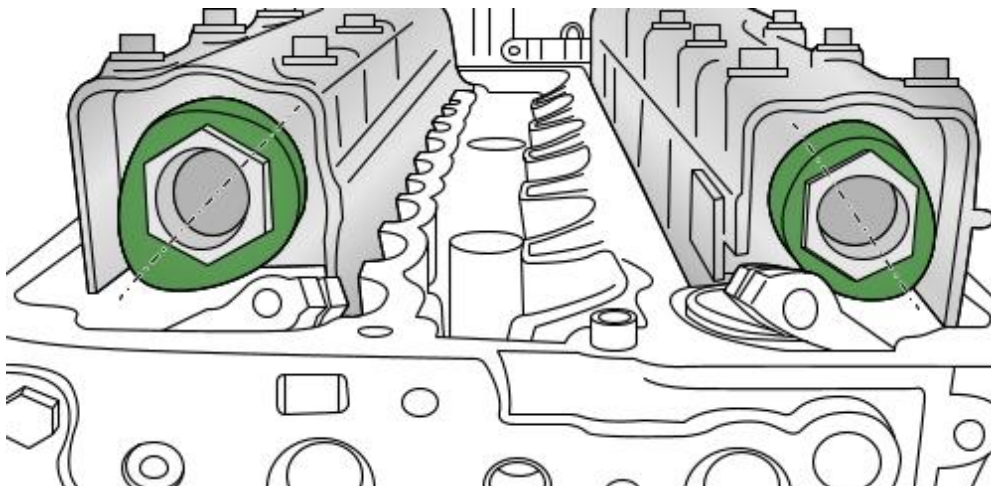
Fit the crankshaft TDC locking pin (11 0 300)

Note: Ensure correct fitting



Check the position of the camshafts

Note: Use a mirror



Fit the camshaft locking tools (11 4 283, 11 4 281)

Check the camshaft gearwheel markings

Exhaust camshaft gearwheel: ((AUS/EX))

Inlet camshaft gearwheel: ((EIN/IN))

Repair Manuals

Refit the camshaft gearwheels

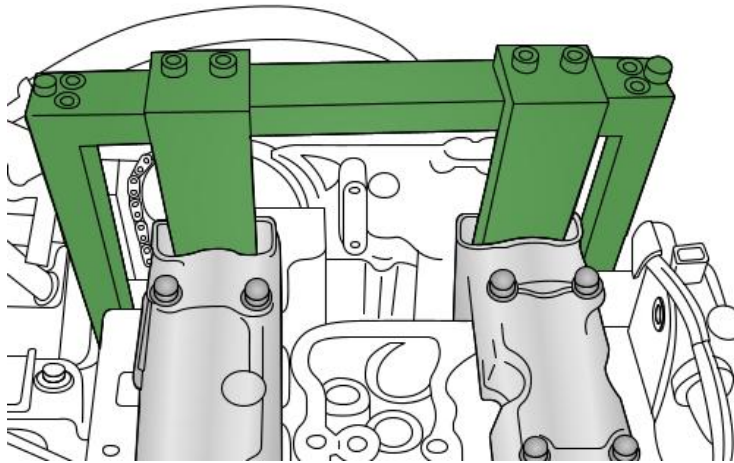
Note: Use new bolts

Adjust the timing

Proceed as follows:

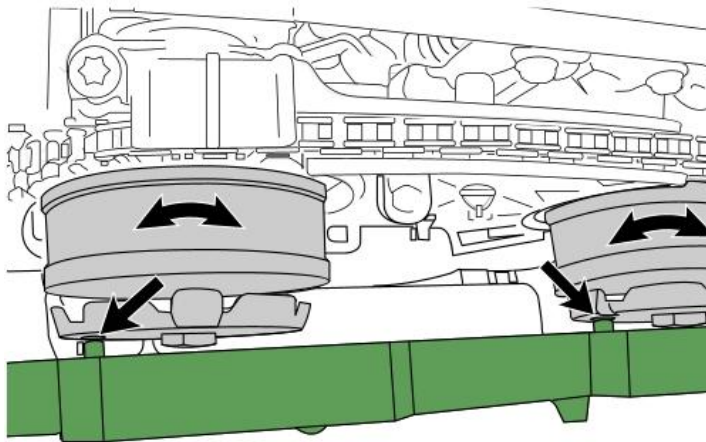
Ensure the crankshaft is locked (11 0 300)

Check that the special tool is correctly fitted (11 4 283, 11 4 281)



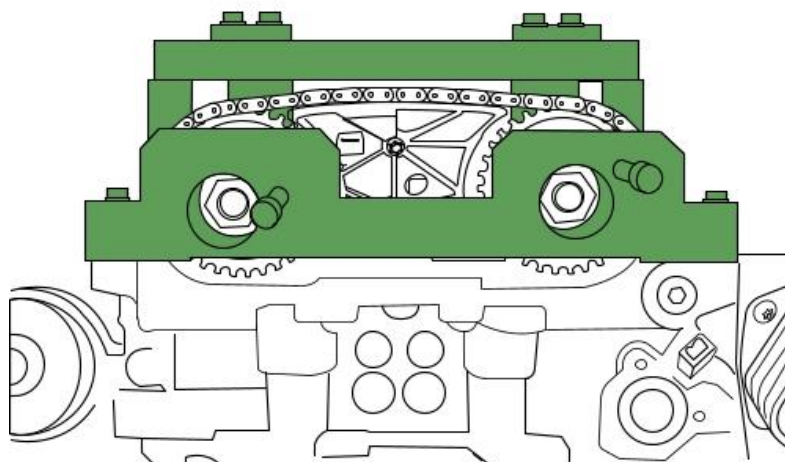
Fit the camshaft alignment tool (11 4 290)

Turn the camshaft gearwheels until the locking tool fits



Secure the special tool with two bolts (11 4 290)

Repair Manuals



Fit the dummy chain tensioner (11 9 340)

Pre-tension the timing chain ((0.6 Nm)) (00 9 250)

Tighten the camshaft gearwheel bolts

Remove the special tools (00 9 250, 11 0 300, 11 4 281, 11 4 282, 11 4 283, 11 4 290, 11 9 340)

Compress the chain tensioner piston twice to drain the oil

Refit the chain tensioner

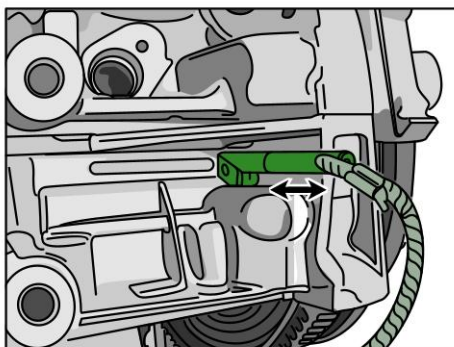
Turn the engine 2 rotations clockwise

Check the valve timing

Proceed as follows:

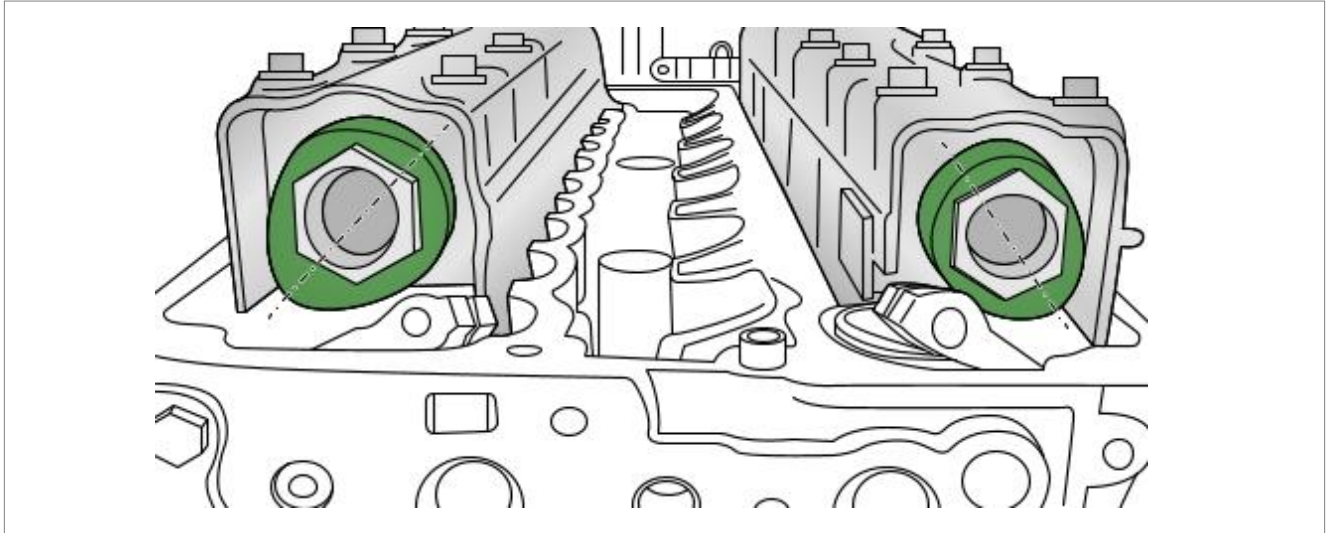
Fit the crankshaft TDC locking pin (11 0 300)

Note: Ensure correct fitting



Check the position of the camshafts

Repair Manuals



Fit the camshaft timing tool (11 4 283, 11 4 281)

If the camshaft locking tool does not fit, proceed as follows:

Loosen the camshaft gearwheel bolts

Note: Use a wrench to counterhold the camshafts

Discard the bolts

Remove the chain tensioner

Adjust the timing

Check the timing again

Further refitting is in the reverse order of removal

Reconnect the battery earth cable

Torque settings

Spark plugs

Spark plugs 23 (Nm)

Injector

Injector 11 - 13 (Nm)

Valve cover

Valve cover 8.5 (Nm)

Timing chain tensioner

Timing chain tensioner (Threads lightly oiled) 55 (Nm)

Renew the seal

Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Ancillary drive belt tensioner

Ancillary drive belt tensioner 40

Camshaft gearwheel

Camshaft gearwheel

Renew the bolts

Stage 1 20 (Nm)

Stage 2 180 (°)

Crankshaft gearwheel

Crankshaft gearwheel (Threads lightly oiled)

Renew the bolts

Stage 1 100 (Nm)

Stage 2 360 (°)

Timing chain guide

Timing chain guide (M6) 8.5 (Nm)

Timing chain guide (M7) 14 (Nm)

Repair Manuals

Timing chain guide (M8) 20 (Nm)

Strut brace:

Use new bolts

M10:

Stage 1: (40 Nm)

Stage 2: (60°)

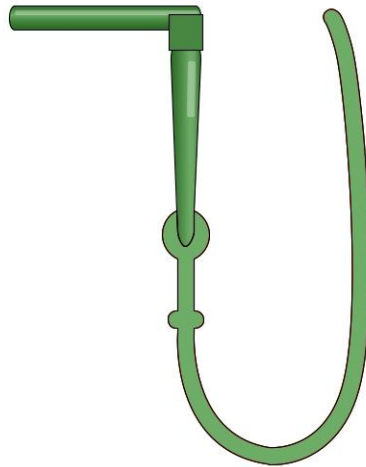
M12:

Stage 1: (100 Nm)

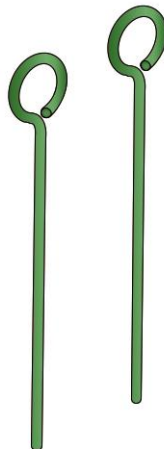
Stage 2: (100°)

Special tools

Crankshaft locking tool: (11 0 300)

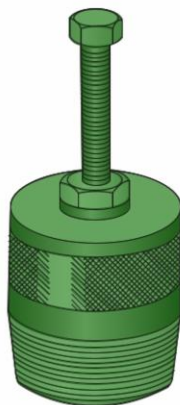


Chain tensioner locking pin: (11 3 340)

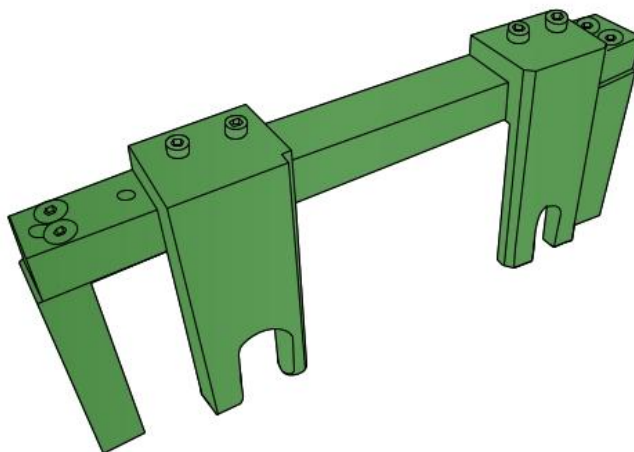


Extractor: (11 0 371, 11 0 372)

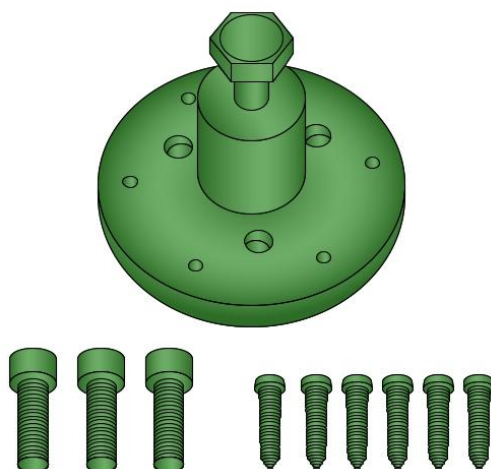
Repair Manuals



Camshaft locking tool: (11 4 283, 11 4 281)

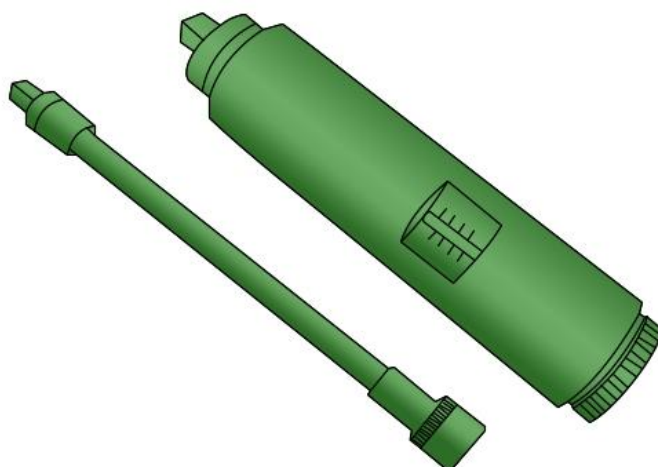


Crankshaft front radial seal fitting tool: (11 9 221, 11 9 222, 11 9 224)

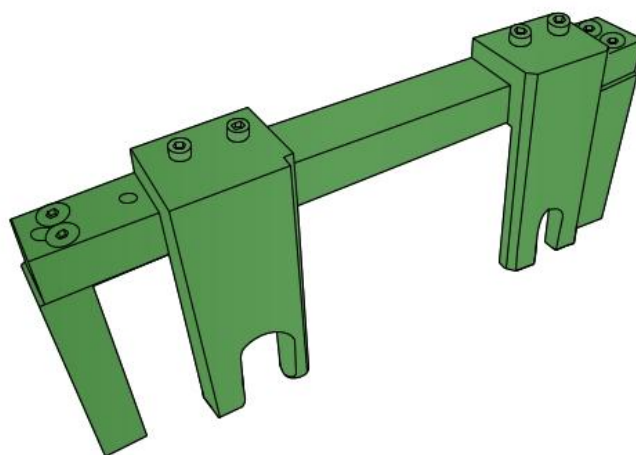


Adjustment tool: (00 9 250)

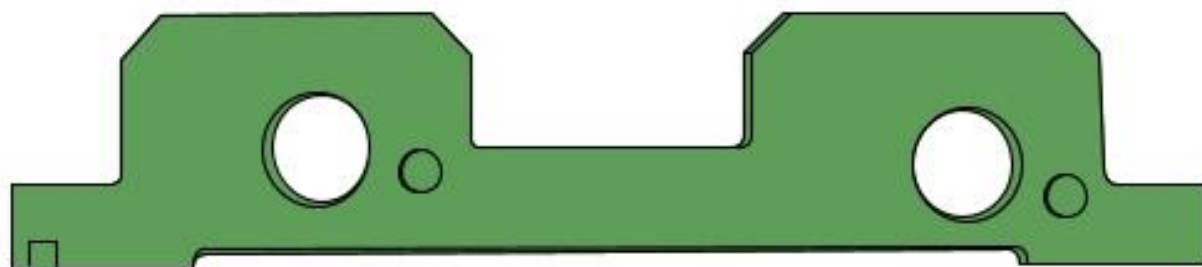
Repair Manuals



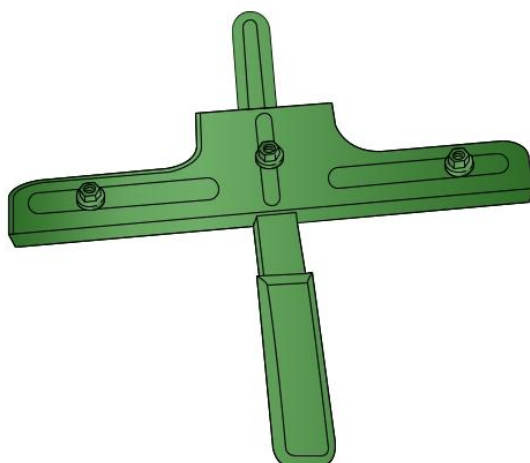
Camshaft locking tool: (11 4 281, 11 4 282, 11 4 283)



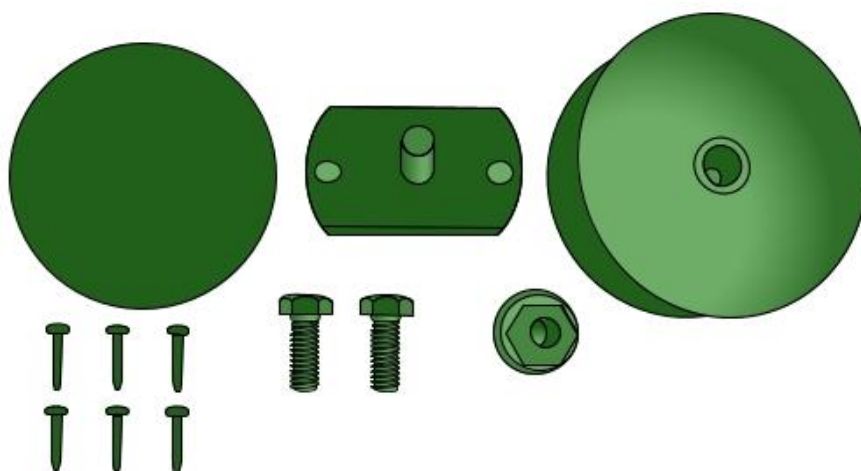
Camshaft alignment tool: (11 4 290)



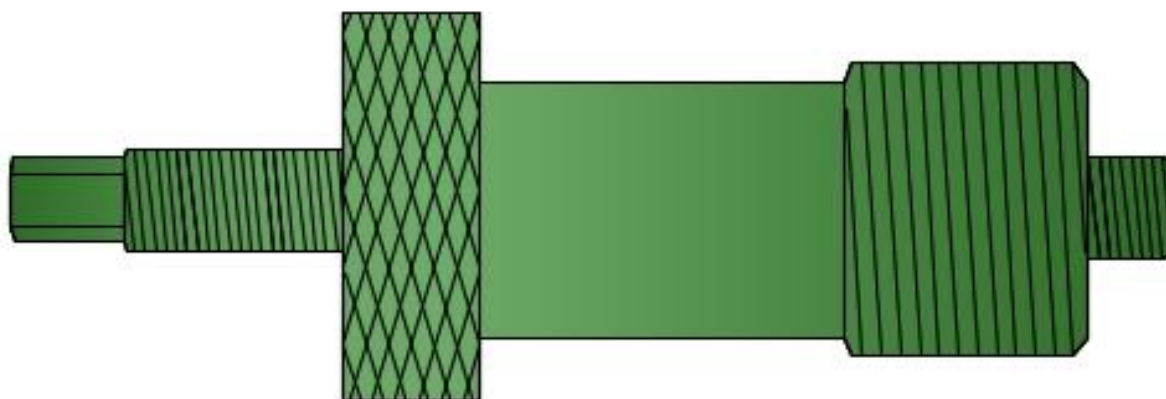
Crankshaft locking tool: (11 8 180)

Repair Manuals

Crankshaft front radial seal fitting tool: (11 9 231, 11 9 232, 11 9 233, 11 9 234)

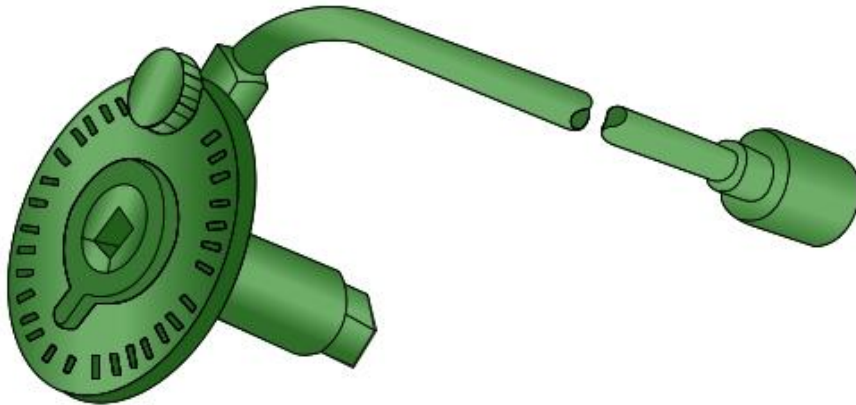


Adaptor: (11 9 340)



Torque wrench: (00 9 140)

Repair Manuals



Oil seal fitting tool: (11 9 232)

